

DIGITAL TRANSFORMATION OF OPERATIONS & SUPPLY CHAIN AT BOSCH INDIA

Proposed AI and Blockchain for Traceability and Quality Assurance

Submitted by

TEAM -6

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1. INTRODUCTION & CONTEXT

India's automakers have experienced a surge in growth driven by increasing demands from within India and outside India, as well as an increase in the use of technology to improve products. With this growth comes the challenge of product tracking, asymmetrical information, and counterfeit parts distributed among numerous levels of their supply chains. The counterfeit automotive parts create risk for the safety of vehicles, brand reputation and corporate bottom line. As one of the largest automotive part- manufacturer to the OEM market in India, Bosch India is now faced with these same compliance problems.

To help mitigate these compliance problems Bosch India has deployed numerous digital initiatives from their Operations and Supply Chain Management departments. This case study illustrates Bosch India's operational digitalisation designs and the development of a full supply chain model using new technologies to improve their supply chains.

Trends:

- **Rapid Growth of Indian Auto Components Industry:** Industry sales are estimated to grow from 6 to 8% in 2026, while auto components revenue to increase by 8- 10%.
- **Digital Transformation in Aftermarket:** IoT, AI- enabled digital twins, and advanced robotics are estimated to reach manufacturing efficiency upwards upto 30%.
- **E-commerce & EV Impact:** The digital online market is estimated at US\$14.3 billion by 2026, with passenger EV to touch from 12-18%.
- **Aftermarket & Consumer Expectations:** The automobile aftermarket is leading towards safety and tech-driven services.
- **National Drive for Traceability & Frontier Technologies:** NITI Aayog warns of a risk of USD 270 billion in manufacturing GDP by 2035 if frontier technologies aren't adopted.
- **Counterfeits hardships at Hand:**
 - **Prevalence & Market Impact:** It has been forecasted that between 20 -25% of the auto part components in the Indian market are, in fact, counterfeits of compromised quality. They also lead to warranty inflation and create brand risk.
 - **Safety Risks & Road Accidents:** Around 20% of the total number of accidents occur due to counterfeit parts.

2. COMPANY BACKGROUND: BOSCH INDIA

Bosch Limited is the Indian Company of the Bosch Group. In other words, Bosch Limited is basically based and operating in India ever since 1951. Its two broad business domains include Mobility Business (Automotive & Auto Components), and Businesses Beyond Mobility (Industrial Technology, Consumer Goods, Building Technologies).

Mob Biz generates ~86% of total revenues; this underscores Bosch India's high reliance on automotive supply chain performance and health. The Bosch India business model is in a position

of technology leadership with high relationships with major OEMs as well as high levels of engineering-driven brand reputation.

Table 1: Key Competitors of Bosch India

Company	Core Strengths	Competitive Position
Bosch India	Advanced technology, OEM trust, strong aftermarket	Market leader, premium positioning
Samvardhana Motherson	Scale, wiring harnesses, global OEM reach	Cost-efficient, high-volume supplier
Sona BLW Precision	Drivetrain & EV components	EV-focused specialist
Continental & ZF India	Global technology & systems	High-end OEM competition
Unorganized / Imports	Low pricing	High counterfeit and quality risk

Pre-Transformation Challenges within the company

- **Siloed Operations:** The state of the manufacturing process, quality process, logistics process, energy process, etc., was in a siloed state. This meant they could not be seen in real time
- **Reactive Maintenance:** Traditionally, the performance of this activity was more on breakdown or schedule-related.
- **Manual Quality Control:** Manual quality control methods were used. Manual methods were basically sampling. It led to an increased level of defects and rework. It became difficult through manual methods to realize the batches in an efficient and speedy way in case of an issue.
- **Rigid Supply Chain:** Weak capabilities in predictive planning, which were unable to react to the volatility in the marketplace concerning supply issues, along with poor traceability in the supply chain.
- **Energy Inefficiency:** Resource usage was assessed periodically instead of being constantly monitored. This impacted efforts towards ensuring the optimization of costs.
- **Digital Skill Gap:** Teams were seen to have poor capabilities in data & analytics functions. As a result, innovation is strongly centered around the capabilities of the central IT functions."

- **Aftermarket risks:** Lack of authenticity led to increased risks of aftermarket issues caused by counterfeit parts and compromised customer trust.

3. BOSCH DIGITAL STRATEGY

The digital transformation program initiated by Bosch India also appears to align quite seamlessly with the global vision of Bosch Group as an organization; to see themselves as a global leader in industry solutions in the mobility business with an incredibly high level of quality, efficiency, and trust with users

Digital Vision:

- Operation Excellences through Industry 4.0
- Enable Data-Driven Decision Making
- Strengthen Quality and Customer Trust
- Support Resilient and Transparent Supply Chains
- Promote Sustainable and Secure Digital Operations

Digital Goals:

- Improve operational efficiency in the area of "manufacturing".
- Improve traceability of the "chain of suppliers" throughout its entire length.
- Improve data-driven decisions with analytics and artificial intelligence.
- Enhance Compliance, Auditability, and Risk Management

Digital Roadmap:

- Phase-1: Knowing Digital: Creating "Data Visibility". Building Internal Digital Infrastructure (ERP-System-Integration, Data-Analytics-for-Quality, IoT-Sensors etc.)
- Phase-2: Doing Digital Applying artificial intelligence, automation, and analytics in optimization strategies.
- Phase-3: Being Digital: This phase is for incorporating the intelligent digital element into daily operations.

Leadership & Governance:

Such a shift is guided by the organizational level leadership to transform the organizational processes through the creation of teams such as sales teams.

It can be sustained through Centres of Excellence and efficient execution styles like speedathons and hackathons.

- Alignment with Business Objectives
- Improves consumer trust
- Strategic alignment with long-term growth and sustainability

- Reduces recalls and enables cost optimization
- Enabling cost and operational efficiency

4. TECHNOLOGIES USED

As part of its digital transformation, Bosch India used the following technologies to realise its VCS.NxT (Value Chain Strategy) and Industry 4.0 ambitions.

Table 2: Technologies used by BOSCH INDIA

Technology	Specific Tools	Purpose & Reason for Choice	Improvements Achieved
Artificial Intelligence (AI) & ML	Supervised/Unsupervised ML, Gen-AI	Used for predictive maintenance, and energy optimization. Provide assistance in everyday processes.	AI-led compressor optimization saved 150+ MWh energy.
Internet of Things (IoT)	Bosch Manufacturing and Logistics Platform (BMLP)	It provides real time machine and production line visibility.	The equipment effectiveness of old machinery increased to 85 %.
Digital Core (ERP)	SAP	Centralizing data for operational excellence and to support digitalization.	Better audit readiness and optimization of IT total costs of ownership.
Intelligent Automation	Robotic Process Automation (RPA), Citizen Developers	Used for workflow digitization. Reduce IT bottlenecks with central team intervention.	Reduced manual touchpoints from 12 to 3 in specific processes; 29% reduction in process time.
Software & Services	Over-the-Air (OTA) Updates, Cloud-Based PCUs	Implement remote diagnostics, and predictive maintenance. Apply the Software-Defined Vehicle trend.	Improved fleet uptime and faster real-time troubleshoot capabilities.

Connectivity Tools	'BeConnected' Digital Tool	To improve retailer engagement and e-warranties.	Onboarded 20,000 retail outlets from 2,000 dealers, which is 2X.
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5. IMPLEMENTATION & CHALLENGES

Implementation Methodology:

- The Bosch India implements the digital initiative in 3 stages using an agile operation governance model. In this connection, it uses 3 stages: “Knowing Digital to Doing Digital to Being Digital”.
- These are then spearheaded by Centres of Excellence for AI, automation, and data analytics for scalable operations.
- Agility is made possible through collaborative partnership: the rapid execution of ideas is done through events like Hackathon and Speed Weeks.
- Digital solutions have been integrated into the BPS GO system for effective implementation of lean principles with data-driven decision-making.

Challenges Encountered:

- Digital Skill gap in employees: With a largely shopfloor-based workforce, there's always a challenge to upscale them with data, analytics, and AI capabilities. This is core for effective digitalization.
- There is a problem with the legacy infrastructure because using machinery that is as old as 25 years and trying to move to Industry 4.0 is difficult. This is because the standards set by Industry 4.0 do require considerable technical resources.
- Cybersecurity risk: With increasing use of IT, data systems, and digital connectivity, the vulnerability of the digital systems to cyber threats and potential manipulations is higher.
- Data silos and quality: To effectively implement digital transformation, clean and consistent data is crucial. Data silos and inconsistency across legacy systems make enterprise level analytics really hard.

Solutions Applied:

- Up skilling of workforce: Bosch has implemented Digital Fluency 2.0 and LEAPS initiatives to develop skills of employees.
- Secure By Design Systems: Products utilize a security-by-design methodology with TISAX and ISO 27001 compliance, with constant penetration tests.

- Digital Infrastructure Modernization: Modern cloud- and artificial intelligence-ready systems are enabled through continuous upgrading of our core IT infrastructure and our dedicated Cyber Defense Center (BCDC).

6. ETHICS, PRIVACY AND SECURITY

Ethics and Compliance:

- There exists a global Compliance Management System based on IDW PS 980. It also renewed its "Code of conduct" in 2024.
- It has a policy of zero tolerance regarding corruption and bribery. It also has a 'Speak Up!' System in place to tackle them.
- The code of conduct of suppliers promotes labor, environment, and human rights compliances throughout the value chain.

Data Protection and Privacy:

- Security framework is compliant with iso 27001 and GDPR. This is managed by a data protection officer
- Guidelines to regulate the full data cycle are also in place for the governance of sensitive data.

Cybersecurity and Risk Management:

- Bosch Cyber Defense Center (BCDC) provides 24x7 monitoring of various threats, while the key plants possess TISAX Level 3 Certification.
- Enterprise risk management utilizes PDCA, and conversely, product risk management utilizes threat analysis and risk assessment.
- It uses "security by design" in its processes and systems. • Mandatory risk & workforce upskilling (Digital fluency 2.0) has been done.

7. BUSINESS IMPACT AND KPI

Table 3: Business Impact & KPI

Business Area	Before Digital Transformation	After Digital Transformation	KPI Impact
Manufacturing & Operations	Limited shopfloor visibility, reactive maintenance, frequent downtime	Real-time monitoring with predictive maintenance and integrated performance dashboards	higher Machine uptime, higher OEE, lower unplanned downtime, lower cycle time
Quality Management	Manual inspections, late defect detection, high rework	Predictive quality control	lower Defect rate (reduced by 17%), higher FPY, lower COPQ, lower scrap and rework
Supply Chain & Planning	Siloed planning and low forecast accuracy. Reactive disruption handling	Data-driven planning with improved logistics	Higher Forecast accuracy, lower order fulfilment lead time, higher inventory turns, 48% reduction in logistics incidents
Sustainability & Resources	Limited emissions visibility with periodic monitoring	Real-time tracking and optimization of utilities	lower energy per unit and better resource efficiency
Decision-Making	Manual reporting, slow, historical insights	Real-time analytics	lower time-to-insight, Faster decision speed
Productivity	High manual effort, fragmented workflows	Digitally enabled workflows	lower manual effort, Higher employee productivity
Customer & Retail Reach	Limited digital engagement with retailers	BeConnected digital platform	Increased each to 20,000 retail outlets

Business Value generated:

- Bosch India had a 8.1 % revenue growth with a stable EBIT of 10.8% in FY 2024-25.
- Exports of the company increased from ₹12,790 million to ₹14,054 million show a push to digital enabled local for global system.

Table 4: BOSCH INDIA Digital Maturity Matrix

Pillar	Maturity Level (1–5)	Current State & Evidence	Strategic Gap
Strategy	Level 4	Digital transformation embedded in Strategy 2030; Industry 4.0, AI, connected products treated as business priorities	Strategy focused internally rather than across the system. Limited platform-based additional services
Technology Infrastructure	Level 4	Actively strengthening the Digital Core and modernizing core systems.	Transitioning 25-year-old legacy machinery as part of digital transformation Systems still partially siloed
Customer Experience	Level 3	The BeConnected platform expanded retail network	Limited anti-counterfeit protection real-time visibility beyond tier 1 suppliers
Operations & Supply Chain	Level 3.9	Executing the VCS.NxT strategy; Industry 4.0 tools used to provide real-time visibility of operations and operational KPI improvements	Agile delivery not yet institutionalized across all functions Lack of deep-tier visibility beyond direct suppliers
Data & Analytics	Level 4	Industrialized over 100 AI and analytics use cases through a dedicated CoE; predictive models reduce internal defect costs and optimize energy.	Ensuring consistent data quality and eliminating silos.

People & Culture	Level 4	Upskilling programs and digital initiative in people through speedathons associate in advanced analytics.	Digital skills uneven across shopfloor.
Governance & Risk	Level 4	Achieved TISAX Level 3 certification. Setup Bosch Cyber Defense Center.	Security and trust mechanisms not extended to aftermarkets.

8. LESSONS AND RECOMMENDATIONS

Lessons learnt:

- Digital transformation not as easy as IT: Bosch's success came from combining the digital tools with the Bosch Production System. The BPS GO! was an accelerator in the digital transformation process.
- Scaling the experimentation: CoEs help Bosch achieve scalability in its investments in AI and automation beyond pilots, hence delivering efficiency, quality, and productivity benefits.
- People and Data are critical enablers: Up-skilling/empowering citizens is digital key for success whereas data silos act as a barrier.
- Resilience and trust are strategic outcomes: Strong governance and cybersecurity measures, such as TISAX Level 3, become key with the expanding digitization across functions.

Recommendations:

Proposed Prototype: “Using AI and Blockchain for Providing a System on Aftermarket Integrity, especially to combat Counterfeit Products”

Create a "proof of concept" permissioned Blockchain that achieves part traceability of a batch in the entire end-to-end supply chain, thus making authenticity verification more secure.

Impact:

- Defect detection lead time - From post complaint to pre dispatch detection.
- Repeat defect rate - measurable reduction across plants, product lines.
- Tier-2/Tier-3 supplier visibility - Significant increase in multi tier transparency.
- Supplier compliance rate - Improved certification and audit adherence.
- Warranty Dispute Resolution Time -Faster and Data-Informed Closure.
- Recall investigation effort - Lower man-hours and cost per incident.

- Counterfeit-linked warranty claims-sustained decline
- OEM trust & satisfaction index - Upward movement.

Others:

- **Ai-Driven Supply Chain Visibility:** Implement an artificial intelligence-based centralized control system to facilitate better management of end-to-end planning, disruption management, and logistics reliability because it will help in identifying what to track batch-wise too.
- **Categorized Enterprise Data Platform:** Take full advantage of information systems associated with the manufacturing, supply chain, customer, and sustainability domains to bridge gaps in information and expedite smooth decision-making processes.
- **Digital Monetization through Services:** A new section would be added that discusses extending product analytics into a subscription service model, including predictive and diagnostic services.

9. CONCLUSION

Bosch India's digital transformation illustrates how emerging digital technologies, including data analytics, artificial intelligence, the Internet of Things, intelligent automation, and enterprise resource planning integration, enable organizations to realize significant gains in areas such as operational efficiency, quality management systems, environmentally sustainable practices, and the ability to analyze information in order to make informed management decisions. By establishing a digital roadmap that clearly separates the steps from Knowing Digital to Being Digital, Bosch India has measured incrementation within its operational Key Performance Indicators (KPI), product quality improvements, and customer experience enhancements.

Despite Bosch India's high internal digital maturity, growing complexity within the automotive supply chain," combined with ongoing issues related to counterfeit products," highlight problems with multi-tier traceability and aftermarket visibility, creating significant risks for both customer safety and brand trust. The use of an Artificial Intelligence (AI) and Blockchain-based Quality Traceability System will allow manufacturers such as Bosch India to fill these gaps by utilizing secure and tamper-proof batch-level traceability of products throughout the entire supply chain." Additionally, by shifting the focus of quality management systems from reactive control measures to proactive risk management, Bosch India can create a superior level of supplier compliance, greater trust from both OEMs and consumers and demonstrate its leadership in the area of digital ecosystems within the manufacturing sector.

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PROTOTYPE LINK

<https://drive.google.com/drive/folders/1GuiVIA7VD9XtWGSzutzrWM5VWsk3OvLu?usp=sharing>

APPENDIX

A. Product-Level Dataset Structure

This section presents the synthetic yet industry -aligned dataset used for product-level authenticity verification in the Bosch spare parts traceability system. Each row represents an individual spare part uniquely identified by a Product ID and linked to its manufacturing batch. The dataset captures critical quality, supplier, and serialization attributes that are commonly used in automotive supply chains to assess part genuineness and risk.

```
import pandas as pd
import hashlib

data = pd.read_csv("C:/Users/SAI NITISH/OneDrive/Desktop/bosch_project2/data/product_quality_data.csv")
data.head()
```

	Product_ID	Batch_ID	Component_Type	Supplier_ID	Inspection_Score	Packaging_Integrity	Serialization_Status
0	P301-01	BATCH_301	Brake_Pad	SUPP_A	92	Yes	Yes
1	P301-02	BATCH_301	Brake_Pad	SUPP_A	90	Yes	Yes
2	P301-03	BATCH_301	Brake_Pad	SUPP_A	88	Yes	Yes
3	P301-04	BATCH_301	Brake_Pad	SUPP_A	91	Yes	Yes
4	P301-05	BATCH_301	Brake_Pad	SUPP_A	89	Yes	Yes

```
data.tail()
```

	Product_ID	Batch_ID	Component_Type	Supplier_ID	Inspection_Score	Packaging_Integrity	Serialization_Status
95	P310-06	BATCH_310	Timing_Belt	SUPP_J	71	No	No
96	P310-07	BATCH_310	Timing_Belt	SUPP_J	77	No	Yes
97	P310-08	BATCH_310	Timing_Belt	SUPP_J	73	No	No
98	P310-09	BATCH_310	Timing_Belt	SUPP_J	76	No	Yes
99	P310-10	BATCH_310	Timing_Belt	SUPP_J	72	No	No

Column explanation:

- Product_ID: Unique identifier for each individual spare part.
- Batch_ID: Manufacturing batch to which the product belongs.
- Component_Type: Type of automotive component (e.g., Brake Pad, Sensor Module).
- Supplier_ID: Tier-level supplier providing the component.
- Inspection_Score: Quality inspection score assigned during manufacturing.
- Packaging_Integrity: Indicates whether original packaging is intact.
- Serialization_Status: Indicates whether the product has valid serialization.

B. Product Authenticity Classification Logic

This section illustrates the Python-based rule engine used to classify each product as Counterfeit, Genuine -High Risk, or Genuine -Low Risk. The classification logic is executed dynamically at runtime and is not stored in the dataset, ensuring integrity and preventing post-verification manipulation.

```
def check_product_authenticity(row):
    if row["Packaging_Integrity"] == "No" or row["Serialization_Status"] == "No":
        return "Counterfeit"
    elif row["Inspection_Score"] < 75:
        return "Genuine - High Risk"
    else:
        return "Genuine - Low Risk"
```

Logic:

- Products with invalid packaging or missing serialization are immediately classified as Counterfeit.
- Products that pass authenticity checks but have low inspection scores are classified as Genuine -High Risk.
- Products with valid packaging, valid serialization, and high inspection scores are classified as Genuine -Low Risk.

C. Blockchain-Inspired Hash Generation Mechanism

This appendix demonstrates the blockchain-inspired hashing mechanism used to generate a tamper-proof digital fingerprint for each product verification event. Instead of storing verification results, a SHA-256 cryptographic hash is generated dynamically using immutable product attributes and the computed authenticity status.

```
def generate_blockchain_hash(row, authenticity_status):

    record = {
        row["Product_ID"]
        + row["Batch_ID"]
        + row["Component_Type"]
        + row["Supplier_ID"]
        + str(row["Inspection_Score"])
        + row["Packaging_Integrity"]
        + row["Serialization_Status"]
        + authenticity_status
    )

    return hashlib.sha256(record.encode()).hexdigest()
```

```
product_id = "P302-04" # change this to test

row = data[data["Product_ID"] == product_id]

if row.empty:
    print("❌ Product not found")
else:
    row = row.iloc[0]

    status = check_product_authenticity(row)
    blockchain_hash = generate_blockchain_hash(row, status)

    print("Product ID:", row["Product_ID"])
    print("Batch ID:", row["Batch_ID"])
    print("Component Type:", row["Component_Type"])
    print("Supplier ID:", row["Supplier_ID"])
    print("Inspection Score:", row["Inspection_Score"])
    print("Packaging Integrity:", row["Packaging_Integrity"])
    print("Serialization Status:", row["Serialization_Status"])
    print("Authenticity Status:", status)
    print("Blockchain Hash:", blockchain_hash)
```

```
Product ID: P302-04
Batch ID: BATCH_302
Component Type: Sensor_Module
Supplier ID: SUPP_B
Inspection Score: 62
Packaging Integrity: No
Serialization Status: No
Authenticity Status: Counterfeit
Blockchain Hash: e40923bb1cd2bcb90800c12c75db9d88e6ee2ce339112f3c0b42a6f9b3ccb395
```

```
for _, row in data.head(5).iterrows():
    status = check_product_authenticity(row)
    hash_value = generate_blockchain_hash(row, status)

    print(row["Product_ID"], "<!--", status)
    print("Hash:", hash_value)
    print("-" * 50)
```

```
P301-01 -> Genuine - Low Risk
Hash: 36a862754d0dbd32d53c5a3bf74d26d2d72dd82d258234904b9b4f37a8412611
-----
P301-02 -> Genuine - Low Risk
Hash: 07de63065e04dcf836afa15acda22a511f8d4caa1d448f1ce90524b0b2c09826
-----
P301-03 -> Genuine - Low Risk
Hash: 287a1fb7df151febb903538e8a5e871b7b029f587d7762de497d3c3503d5acd1
-----
P301-04 -> Genuine - Low Risk
Hash: d01b96c306239fc2ba80bb95b13fb61a8fadb7398b5bf7f0148d65ebc9fea49
-----
P301-05 -> Genuine - Low Risk
Hash: 7dbc5d5d537e57f8a10105d862686da2534c609ae737d5d3169552eabe581075
-----
```

```
for _, row in data.tail(5).iterrows():
    status = check_product_authenticity(row)
    hash_value = generate_blockchain_hash(row, status)

    print(row["Product_ID"], "<!--", status)
    print("Hash:", hash_value)
    print("-" * 50)
```

```
P310-06 -> Counterfeit
Hash: 865a36c7af3017ceaf9712b3d165ed43640b1b5f447c89667424739ed3880172
-----
P310-07 -> Counterfeit
Hash: 0d926cc5c28524e0ad84d6d4a90d601df3a880c1cd95afae5b42cc6fbd00d341
-----
P310-08 -> Counterfeit
Hash: 33b061d55b88d2604402ff5a32865a91da8d8a6df195ecaecca08b333042769b
-----
P310-09 -> Counterfeit
Hash: 82b276eb2bada6f22938789f52a08fd49f6345975e6722d744b833647b13d65
-----
P310-10 -> Counterfeit
Hash: e4acd8c35c5b5d14f20f715b0d6282b26aee3ac4c23be308807b59b115bdafae9
-----
```

NOTE: Any change in product attributes or verification outcome results in a completely different hash, thereby simulating blockchain immutability and ensuring end-to-end traceability without requiring a full distributed ledger implementation.

D. Product-Level Authenticity Verification Results

This section demonstrates the working output of the Product-Level Spare Part Authenticity Verification Portal, developed using Python and Streamlit.

This objective is to show how individual products (not batches) are classified into three risk outcomes using operational and quality parameters, along with a blockchain-style verification hash.

Verification Logic Overview

Each product is evaluated using the following three key checks

- Packaging Integrity
- Serialization Status
- Inspection Score

Based on these inputs, the system classifies products into:

- Genuine -Low Risk
- Genuine -High Risk
- Counterfeit

CASE -1: Counterfeit Product Detection

Tested Product

- Product ID: P310-10
- Batch ID: BATCH_310
- Component Type: Timing_Belt

Observed Conditions

- Packaging Integrity: No
- Serialization Status: No
- Inspection Score: 72

System Output

- Risk Classification: Counterfeit
- Blockchain Verification Hash: Generated and displayed

Spare Part Authenticity Verification Portal

Enter the Product ID to check whether the product is genuine, high risk, or counterfeit.

Enter Product ID (e.g., P301-01)

P310-10

Verify Product

✗ Product is COUNTERFEIT

Product Details

Product ID: P310-10

Batch ID: BATCH_310

Component Type: Timing_Belt

Supplier ID: SUPP_J

Inspection Score: 72

Packaging Integrity: No

Serialization Status: No

Risk Classification: Counterfeit

Blockchain Verification Proof

a19b0c80dc1c48b19e32cb5c64bb94d7f2971d330805ce7306b48540f8b51f28

Interpretation

The system flags this product as Counterfeit because:

- Either packaging integrity or serialization status is compromised
- These two checks are treated as mandatory authenticity controls
- Even though an inspection score exists, failure of these controls indicates high likelihood of counterfeit entry

This reflects real-world supply chain logic, where missing serialization or damaged packaging invalidates authenticity.

CASE -2: Genuine but High-Risk Product

Tested Product

- Product ID: P302-02
- Batch ID: BATCH_302
- Component Type: Sensor_Module

Observed Conditions

- Packaging Integrity: Yes
- Serialization Status: Yes
- Inspection Score: 68

System Output

- Risk Classification: Genuine -High Risk
- Blockchain Verification Hash: Generated and displayed

Spare Part Authenticity Verification Portal

Enter the Product ID to check whether the product is genuine, high risk, or counterfeit.

Enter Product ID (e.g., P301-01)

P302-02

Verify Product

⚠️ Product is GENUINE but HIGH RISK

Product Details

Product ID: P302-02

Batch ID: BATCH_302

Component Type: Sensor_Module

Supplier ID: SUPP_B

Inspection Score: 68

Packaging Integrity: Yes

Serialization Status: Yes

Risk Classification: Genuine - High Risk

Blockchain Verification Proof

9d8aa508c177ee5e4b94710e5a54781e859c0e4d4a68aa48937d4fbb08b3bc7a

Interpretation

The product is considered genuine, but:

- The inspection score is below the safe threshold (75)
- This suggests quality deviation or potential performance risk
- Product should not be immediately rejected, but requires:
 - Additional Inspection
 - Controlled release
 - Warranty or recall monitoring

This category is crucial in quality-sensitive industries like automotive manufacturing.

CASE -3: Genuine and Low-Risk Product

Tested Product

- Product ID: P301-01
- Batch ID: BATCH_301
- Component Type: Brake_Pad

Observed Conditions

- Packaging Integrity: Yes

- Serialization Status: Yes
- Inspection Score: 92

System Output

- Risk Classification: Genuine -Low Risk
- Blockchain Verification Hash: Generated and displayed

Spare Part Authenticity Verification Portal

Enter the Product ID to check whether the product is genuine, high risk, or counterfeit.

Enter Product ID (e.g., P301-01)

P301-01

Verify Product

✔ Product is GENUINE and LOW RISK

Product Details

Product ID: P301-01
Batch ID: BATCH_301
Component Type: Brake_Pad
Supplier ID: SUPP_A
Inspection Score: 92
Packaging Integrity: Yes
Serialization Status: Yes
Risk Classification: Genuine - Low Risk

Blockchain Verification Proof

e7b3cbbe45c3a2796f02090afcb71a05856ff0f05232891d7d691a5c683a6723

Interpretation

This product satisfies all authenticity and quality conditions:

- Secure packaging
- Valid serialization
- High inspection score

It is safe for direct distribution and customer use, with minimal operational risk.